Lab 06 Report

Write a short report (3-5 paragraphs) that summarizes the evidence from the trial for adopting each of the interventions or not. Use tables and visualizations. Write in a professional tone for a technical audience, and focus on evidence (or the lack thereof). Do not assume anyone has read your work in your notebooks.

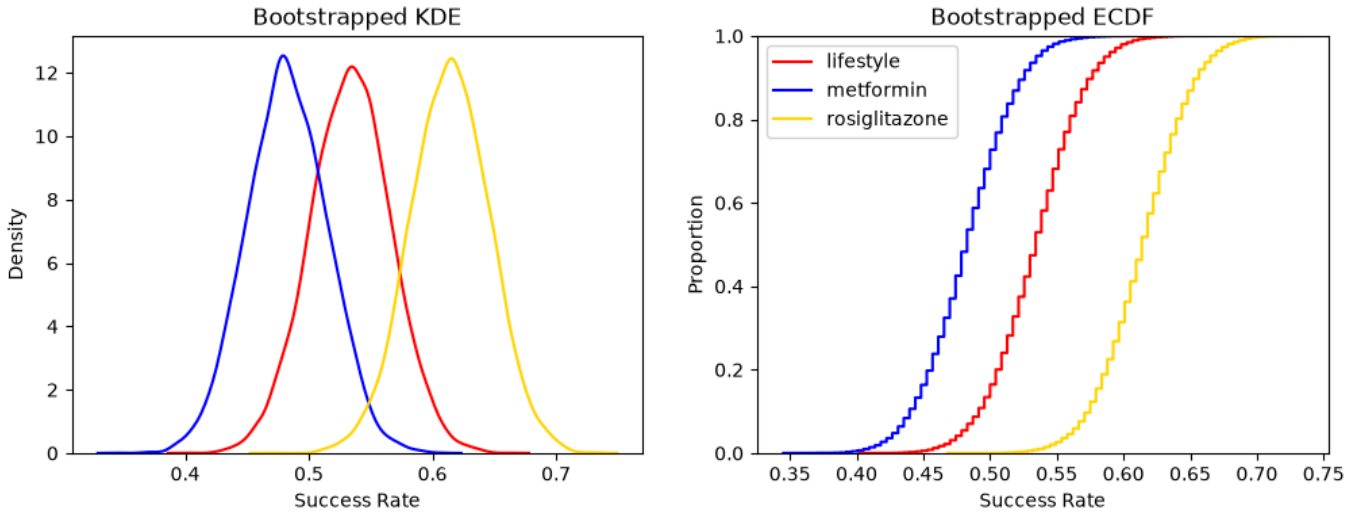


Figure 1: The KDE and ECDF of boostrapped samples of the success rates of three different treatment strategies for pre-diabetes.

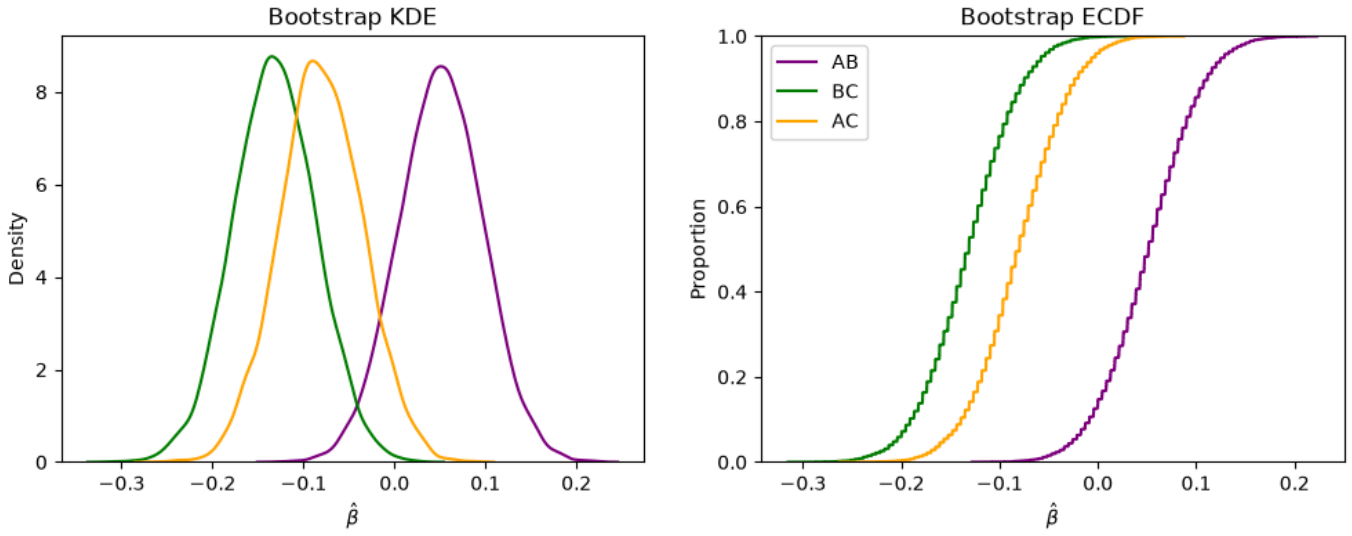


Figure 2: The KDE and ECDF of boostrapped samples of ATE (Average Treatment Effect) between every combination of the three different treatment strategies for pre-diabetes.

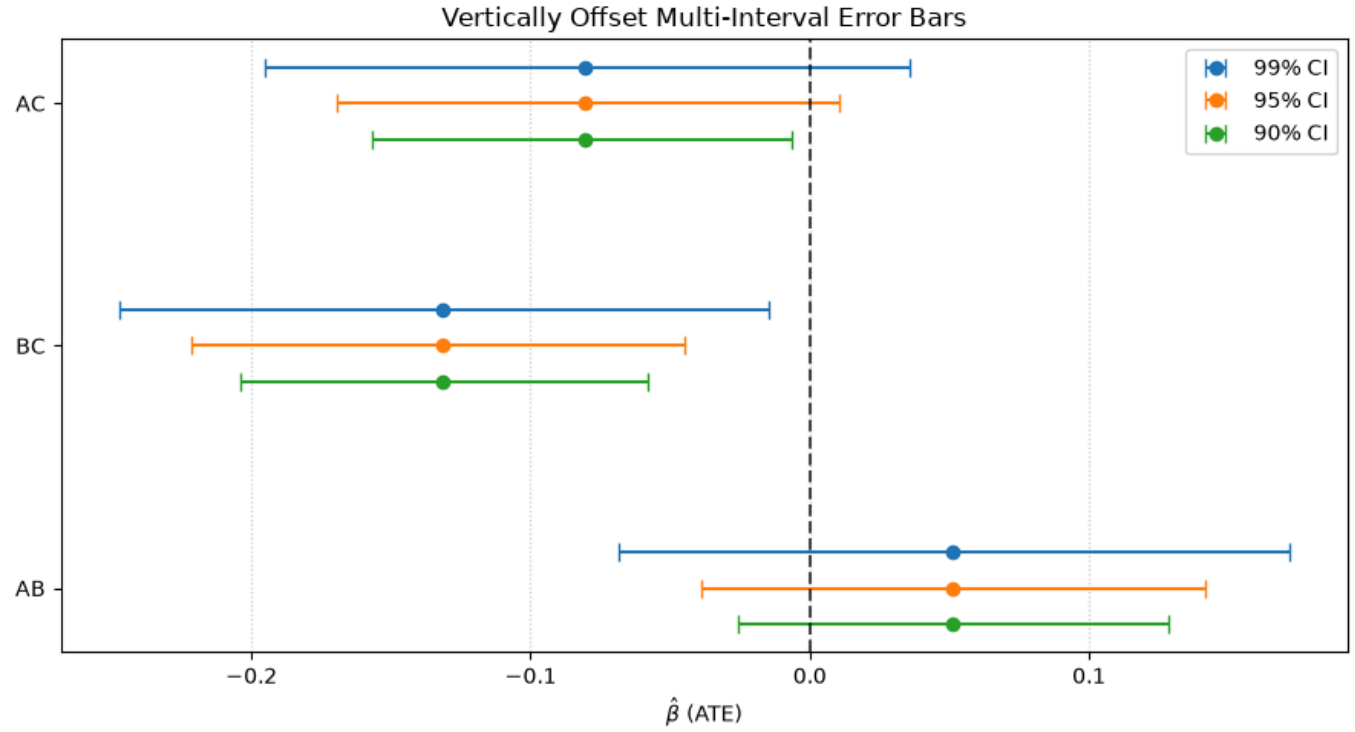


Figure 3: The 90%, 95%, and 99% confidence intervals of the difference in ATEs for each combination of treatment strategies for pre-diabetes.

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	failure	success	failure_proportion	success_proportion
lifestyle	**109**	**125**	**0.47**	**0.53**
met	**120**	**112**	**0.52**	**0.48**
rosi	**90**	**143**	**0.39**	**0.61**

Table 1: Table showing count and proportion of successes and failures across the 3 treatments

Intervention A (Lifestyle)

Intervention A has a positive success rate centered around 53% according to Figure 1 and Table 1. The ECDF of Intervention A is dominated by Intervention C in Figure 1 while Intervention A dominates Intervention B. In Figure 2, the ECDF of the difference in ATE for Interventions A and B and the difference in ATE for Interventions A and C dominates the difference in ATE between B and C.

When comparing Intervention A to Interventions B and C, only the 90% confidence interval for the ATE difference between A and C doesn't include zero. This means that when considering treatment A, in most cases, we cannot say that there is a difference in choosing another treatment. The only plausible difference would be between Intervention A and Intervention C within the 90% confidence interval.

Applicatively, this means that when exploring treatment options for pre-diabetes, one should consider other factors outside of success rates such as cost, medication side-effects, medication availability, etc. because there is not an appreciatable difference in the average treatment effects between lifestyle changes and the Rosiglitazone treatment or lifestyle changes and the Metformin treatment.

Intervention B (Metformin)

According to Figure 1 and Graph 1, the Metformin average success rate is around 48%. This means that less than half of the time, the Metformin drug is successful. The ECDF of Intervention B is dominated by both Interventions A and C in Figure 1. In Figure 2, the dominant ECDF is the difference in ATE between Interventions A and B.

When comparing Intervention B to Intervention A, there is no discernable difference at any of the confidence intervals in Figure 3. There is, however, a discernable difference between Interventions B and C as zero is not captured within any of the confidence intervals. Additionally, the difference between ATEs for Interventions B and C is consistently negative. This means that there is evidence to support that there is a difference between Interventions B and C and that, specifically, C is a more effective treatment.

Intervention C (Rosiglitazone)

According to Figure 1 and Table 1, Intervention C dominates both other treatments with a 61% success rate. All combinations of ATEs have been covered in previous sections of this report, but just to reiterate the only significant result in ATE difference was between Intervention B and Intervention C as seen in Figure 3.

In practice, if one was taking Metformin, switching to taking Rosiglitazone results in the largest and most significant increase in effects.